CHEM 1301 Driller 9-4

Section 8.7 (empirical and molecular formulas) + Chapter 9 — 15 exam-style multiple-choice questions

Instructions. Choose the single best answer (a–e) for each question. You may use a periodic table. Calculators are allowed. Answers are rounded to the precision shown. Time yourself: aim for about 30 minutes (roughly 2 minutes per question, the same pace as the full exam).

Answer sheet. Use the bubble answer sheet (`CHEM1301/exam-prep/answer-sheet/answer-sheet.pdf`), rows 1–15.

1. While finding an empirical formula, you divide by the smallest mole value and get C 1.00, H 1.33, O 1.00. What should you do next?

a) Divide every result by 1.33 again
b) Multiply every element's result by 3, giving $C_3H_4O_3$
c) Multiply every result by 2, giving $C_2H_3O_2$
d) Multiply only the H result by 3, giving CH_4O
e) Round H to 1, giving CHO

2. Erythrose has the molecular formula $C_4H_8O_4$. What is its empirical formula?

a) CH_4O
b) $C_4H_8O_4$
c) CHO
d) CH_2O
e) $C_2H_4O_2$

3. Aspirin is 60.00% C, 4.48% H, and 35.52% O by mass. What is its empirical formula?

a) C_2H_2O
b) $C_{18}H_{16}O_8$
c) C_9H_2O
d) $C_{13}HO_8$
e) $C_9H_8O_4$

4. An orange salt is 26.58% K, 35.35% Cr, and 38.07% O by mass. What is its empirical formula?

a) $K_2Cr_2O_7$
b) KCrO
c) $KCrO_7$
d) $KCrO_4$
e) $K_4Cr_4O_{14}$

5. A sample of a sodium salt contains 2.30 g of Na, 1.60 g of S, and 2.40 g of O. What is its empirical formula?

a) Na_3SO_2
b) NaSO
c) $Na_4S_2O_6$
d) Na_2SO_3
e) Na_3SO_3

6. A hydrocarbon has the empirical formula C_3H_4 and a molar mass of 120.19 g/mol. What is its molecular formula?

a) C_6H_8
b) $C_{12}H_{16}$
c) C_9H_4
d) C_3H_4
e) C_9H_{12}

7. A liquid fuel is 52.14% C, 13.13% H, and 34.73% O by mass, and its molar mass is 46.07 g/mol. It burns completely in O_2 to form CO_2 and H_2O. How many moles of O_2 are needed to burn 0.200 mol of the fuel?

a) 0.200 mol
b) 0.400 mol
c) 0.600 mol
d) 0.700 mol
e) 1.20 mol

8. To convert **moles of A** into **moles of B** for a reaction, which information do you need?

a) The coefficients of A and B in the balanced equation
b) Avogadro's number
c) The masses of A and B present
d) The molar masses of A and B
e) The subscripts in the formulas of A and B

9. For $4NH_3(g) + 3O_2(g) \rightarrow 2N_2(g) + 6H_2O(g)$, how many moles of N_2 form from 2.40 mol of NH_3?

a) 1.20 mol
b) 1.80 mol
c) 2.40 mol
d) 4.80 mol
e) 33.6 mol

10. For $CS_2(l) + 3O_2(g) \rightarrow CO_2(g) + 2SO_2(g)$, how many moles of SO_2 form from 15.2 g of CS_2 with excess O_2?

a) 0.0998 mol
b) 0.200 mol
c) 0.399 mol
d) 0.599 mol
e) 25.6 mol

11. Copper displaces silver: $Cu(s) + 2AgNO_3(aq) \rightarrow Cu(NO_3)_2(aq) + 2Ag(s)$. What mass of Ag forms when 5.00 g of Cu reacts with excess $AgNO_3$?

a) 0.157 g
b) 4.24 g
c) 8.49 g
d) 10.0 g
e) 17.0 g

12. Potassium burns according to the **unbalanced** equation $K(s) + O_2(g) \rightarrow K_2O(s)$. What mass of K_2O forms from 7.82 g of K with excess O_2?

a) 0.100 g
b) 3.91 g
c) 4.71 g
d) 9.42 g
e) 18.8 g

13. Zinc sulfide is roasted: $2ZnS(s) + 3O_2(g) \rightarrow 2ZnO(s) + 2SO_2(g)$. A mixture of 50.0 g of ZnS and 20.0 g of O_2 reacts. Which statement is correct?

a) ZnS is limiting, and O_2 is left over
b) O_2 is limiting, and 9.40 g of ZnS is left over
c) O_2 is limiting, and 50.0 g of ZnS is left over
d) O_2 is limiting, and 40.6 g of ZnS is left over
e) Neither is limiting; both reactants are used up completely

14. For $2Al(s) + 3CuSO_4(aq) \rightarrow Al_2(SO_4)_3(aq) + 3Cu(s)$, 5.40 g of Al reacts with 40.0 g of $CuSO_4$. What is the theoretical yield of Cu?

a) 0.251 g
b) 15.9 g
c) 19.1 g
d) 40.0 g
e) 45.4 g

15. Magnesium burns: $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$. Burning 4.86 g of Mg in excess O_2 gives 7.10 g of MgO. What is the percent yield?

a) 0.881%
b) 17.6%
c) 88.1%
d) 114%
e) 146%